

Blind Calibration Method of Gain and Time-skew Mismatches in Time-interleaved ADCs

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Abstract—This paper proposes a novel digital blind calibration method to calibrate the gain and time-skew mismatches in any-channel time-interleaved analog-to-digital converters (TI-ADCs). Based on the Hilbert transformation and derivative operation, the spurious signals due to the gain and time-skew mismatches could be reconstructed and subtracted from the sampled signal blindly. The proposed calibration method is applicable to any channel TI-ADC with no limitations to input signal type. Extensive simulations results have been presented to demonstrate the effectiveness of the proposed method, which shows that the proposed calibration method could improve the SNDR and SFDR by 30 dB and 45 dB respectively.

I. INTRODUCTION

The time-interleaved analog-to-digital converters (TI-ADCs) are widely used in the high speed analog-to-digital conversion since the newer CMOS technologies no longer provide significant speed advantages for a monolithic ADC [1]. The TI-ADC architecture employs parallel sub-ADCs to sample the analog signal cyclically at different time instances, which could improve the sampling rate by M times for an M-channel TI-ADC theoretically. However, the main drawback of TI-ADC is that the offset, gain and time skew mismatches among sub-ADCs cause spurious spectral components, which would significantly degrade the signal quality, i.e., poor ENOB and SNDR [2]. Thus, the key to a high performance solution is to eliminate the mismatches among the sub-ADCs.

The researchers have proposed numerous digital calibration methods for the TI-ADCs in the last decade. The main focus is on the calibration of offset, gain and timing skew mismatches [3]. The offset mismatches cause several spurs spaced at the fixed frequency (i.e. $2k\pi/M$), which could be eliminated with a simple notch filter or an average block. Thus the most urgent problem is to find the effective calibration method for gain and time-skew mismatches, which could further improve the performance of the TI-ADC systems [4]–[6]. In [4], the authors proposed a digital blind calibration method based on Hadamard transform and the derivative operation. However, this method only adapts to the TI-ADCs with specific channel numbers with corresponding Hadamard transforms. Also, the performance of this method significantly degrades when the input signal approaches the Nyquist frequency. [5] presented a digital calibration method for any channel TI-ADCs based on

polyphase decomposition. In this calibration structure, each channel requires an individual SGD feedback system and the estimation accuracy degrades when the time-skews are large. In [6], a low complexity digital calibration method for timing mismatch is proposed. The author took the first channel as the reference one and the other channels are calibrated in turns. However, this method only adapts to the channel number 2^N and the calibration structure will become very complicated when the channel number increases.

This paper proposes a blind calibration method of gain and time-skew mismatches in TI-ADC systems. It does not require any advance information of the input signal and the adaptive filter banks are also avoided. The basis functions of gain and time-skews are generated from the Hilbert transformation and derivative operation with two simple FIR filters. Then the mismatch coefficients could be estimated based on the least square (LS) algorithm. Specially, this method is applicable to any channel TI-ADC without loss at high frequency.

II. PROPOSED BLIND CALIBRATION METHOD

A. System Model of TI-ADC

In this subsection, the basic system model of TI-ADC is introduced first. Then the relationship between gain and time-skew mismatches and the spurious signals is analyzed. At last, the blind calibration theory is established.

A block diagram of an M channel TI-ADC with gain and time-skew mismatches is shown in Fig. 1. Each channel consists of a sub-ADC, whose sampling frequency is F_s/M . Then, the overall sampling frequency of the TI-ADC is F_s . The discrete-time Fourier transform (DTFT) of the k-th down-sampled signal could be written as [3]

$$Y_k(j\omega) = e^{-jk\omega} \frac{1}{M} \sum_{i=0}^{M-1} H_k \left(j \left(\omega - \frac{2\pi i}{M} \right) \right) X \left(j \left(\omega - \frac{2\pi i}{M} \right) \right) \quad (1)$$

with

$$H_k(j\omega) = (1 + \Delta_{gk}) e^{j\omega(\Delta_{tk} + \Delta_{gk})}, k = 0, 1, \dots, M-1 \quad (2)$$

where Δ_{gk} and Δ_{tk} donate the k-th channel gain and time-skew mismatch. In an ideal TI-ADC, these mismatch values are

identical for all the sub-channels. The overall output of the TI-ADC could be simplified as

$$Y(j\omega) = \sum_{k=0}^{M-1} Y_k(j\omega) = X(j\omega) + \sum_{i=1}^{M-1} S_i(j\omega) \quad (3)$$

where $X(j\omega)$ represents the desired signal and $S_i(j\omega)$ is the i -th spurious signals. In the real applications, the gain and time-skew mismatches are usually much less than one. Thus the spurious signals could be further modified as

$$S_i(j\omega) \approx w_{gi} X\left(j\left(\omega - \frac{2\pi i}{M}\right)\right) + w_{ti} \hat{X}\left(j\left(\omega - \frac{2\pi i}{M}\right)\right) \quad (4)$$

with

$$\begin{aligned} \hat{X}(j\omega) &= j\omega X(j\omega) \\ w_{gi} &= \frac{1}{M} \sum_{k=0}^{M-1} e^{-\frac{2\pi i k j}{M}} \Delta_{gk} \\ w_{ti} &= \frac{1}{M} \sum_{k=0}^{M-1} e^{-\frac{2\pi i k j}{M}} \Delta_{tk} \end{aligned} \quad (5)$$

where w_{gi} and w_{ti} are the mismatch coefficient of the gain and time-skew respectively.

Since the desired signal $X(j\omega)$ is unknown in a blind calibration method, the spurious signals could not be directly constructed according to (4). Luckily, the power of spurious signals is usually much lower than the desired signal. Therefore, it is reasonable to replace $X(j\omega)$ with $Y(j\omega)$ in (4), which leads to the form of the calibrated output signal as

$$\begin{aligned} Y_c(j\omega) &= Y(j\omega) - \sum_{i=1}^{M-1} \tilde{w}_{gi} Y\left(j\left(\omega - \frac{2\pi i}{M}\right)\right) \\ &\quad - \sum_{i=1}^{M-1} \tilde{w}_{ti} \hat{Y}\left(j\left(\omega - \frac{2\pi i}{M}\right)\right) \end{aligned} \quad (6)$$

where $\tilde{w}_{gi}$ and $\tilde{w}_{ti}$ are the estimation value of w_{gi} and w_{ti} . The next subsection will introduce the estimation of mismatch coefficients.

B. Estimation of mismatch coefficients

It is noticed that the above equations are the frequency-domain expression. By finding the IDFT of (6), the time-domain expression of the calibrated signal could be written as

$$y_c[n] = y[n] - \sum_{i=1}^{M-1} \tilde{w}_{gi} y_i[n] - \sum_{i=1}^{M-1} \tilde{w}_{ti} \hat{y}_i[n] \quad (7)$$

with

$$\begin{aligned} y_i[n] &= HT\left\{\text{real}\left(y[n] \cdot e^{2\pi i n j / M}\right)\right\} \\ \hat{y}_i[n] &= HT\left\{\text{real}\left((y[n] * h_D[n]) \cdot e^{2\pi i n j / M}\right)\right\} \end{aligned} \quad (8)$$

where $h_D[n]$ donates the coefficients of the derivative filter and $HT\{\cdot\}$ indicates the Hilbert transform. Based on the LS algorithm, the mismatch coefficients could be calculated as the matrix form

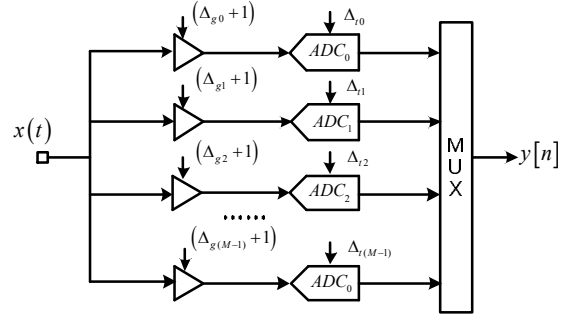


Fig. 1. The model of an M-channel TI-ADC.

$$\begin{aligned} \mathbf{w}_g &= (\mathbf{B}_g^T \mathbf{B}_g)^{-1} \mathbf{B}_g^T \mathbf{y} \\ \mathbf{w}_t &= (\mathbf{B}_t^T \mathbf{B}_t)^{-1} \mathbf{B}_t^T \mathbf{y} \end{aligned} \quad (9)$$

with

$$\begin{aligned} \mathbf{w}_g &= [\tilde{w}_{g1} \quad \tilde{w}_{g2} \quad \dots \quad \tilde{w}_{g(M-1)}]^T \\ \mathbf{w}_t &= [\tilde{w}_{t1} \quad \tilde{w}_{t2} \quad \dots \quad \tilde{w}_{t(M-1)}]^T \\ \mathbf{y} &= [y[n+1] \quad y[n+2] \quad \dots \quad y_1[n+N]]^T \\ \mathbf{B}_g &= \begin{bmatrix} y_1[n+1] & y_2[n+1] & \dots & y_{M-1}[n+1] \\ y_1[n+2] & y_2[n+2] & \dots & y_{M-1}[n+2] \\ \dots & \dots & \dots & \dots \\ y_1[n+N] & y_2[n+N] & \dots & y_{M-1}[n+N] \end{bmatrix} \\ \mathbf{B}_t &= \begin{bmatrix} \hat{y}_1[n+1] & \hat{y}_2[n+1] & \dots & \hat{y}_{M-1}[n+1] \\ \hat{y}_1[n+2] & \hat{y}_2[n+2] & \dots & \hat{y}_{M-1}[n+2] \\ \dots & \dots & \dots & \dots \\ \hat{y}_1[n+N] & \hat{y}_2[n+N] & \dots & \hat{y}_{M-1}[n+N] \end{bmatrix} \end{aligned} \quad (10)$$

Actually, the $\mathbf{B}_g$ and $\mathbf{B}_t$ are the basis-function matrixes of the spurious signals due to gain and time-skew mismatches respectively. The mismatch coefficients could be estimated by calculating the pseudo-inverse operation of the basis-function matrixes. The method could also be designed as an adaptive estimation manner based on the LMS or RLS algorithm by calculating the correlation between the basis functions and the sampled signal [3]-[6].

III. SIMULATION RESULTS

In this section, the extensive simulations are presented to demonstrate the effectiveness of the proposed blind calibration method. The simulated TI-ADC model is based on Fig. 1 and the resolution of the sub-ADC is 10 bit. To illustrate the extendability of the proposed calibration method, a 4-channel TI-ADC and 16-channel TI-ADC is simulated respectively.

A. Simulation of 4-channel TI-ADC

Based on (2), the values of $\Delta_{g0} \sim \Delta_{g3}$ were chosen as 0.03, 0.02, -0.03 and -0.04. The value of $\Delta_{t0} \sim \Delta_{t3}$ were set as 0.04,

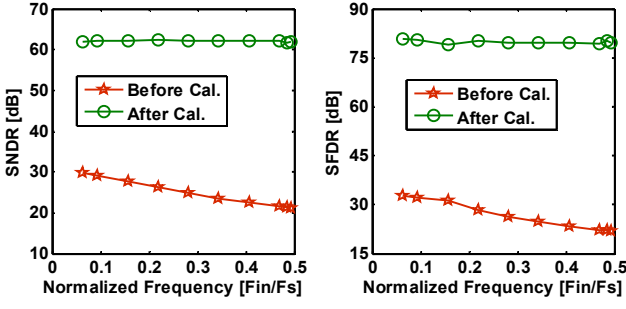


Fig. 2. SNDR (left) and SFDR (right) performance versus the input frequency for a 4-channel TI-ADC.

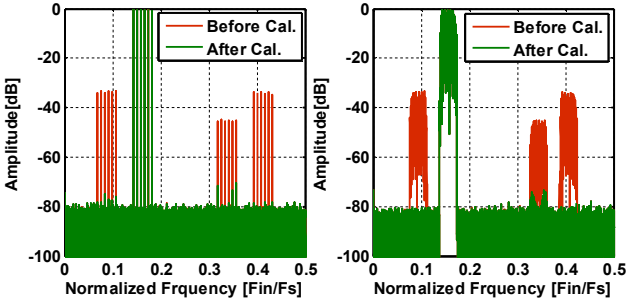


Fig. 3. Spectra of a 4-channel TI-ADC for multi-tone (left) and 16QAM (right).

-0.03, 0.02 and -0.02. Firstly, we fed a single tone input whose frequency varies from 0 to $0.5 F_s$ to the 4-channel TI-ADC. The SNDR and SFDR of the sampled signal before/after calibration are depicted in Fig. 2. The proposed calibration method could improve the SNDR and SFDR by 30 dB and 45 dB respectively in the first Nyquist zone.

In addition, the multi-tone and wideband (16QAM) signal were tested under the proposed calibration method as shown in Fig. 3. Before calibration, the SNDR / SFDR of the multi-tone and 16QAM are 29.86 dB / 33.28 dB and 30.15 dB and 33.61 dB respectively. After calibration, the SNDR / SFDR of the multi-tone and 16QAM are 61.17 dB / 70.43 dB and 61.98 dB / 72.27 dB respectively. The proposed calibration method could effectively improve the SNDR and SFDR performance for the 4-channel TI-ADC system.

B. Simulation of 16-channel TI-ADC

Furthermore, a 16-channel TI-ADC is simulated similarly. The values of $\Delta_{g0} \sim \Delta_{g15}$ were set as 0.045, 0.035, -0.04, -0.025, 0.045, 0.03, -0.04, -0.015, 0.03, 0.035, -0.025, -0.035, 0.03, 0.03, -0.04, and -0.025. The value of $\Delta_{t0} \sim \Delta_{t15}$ were set as 0.03, -0.015, 0.02, -0.01, 0.015, -0.02, 0.015, -0.02, 0.01, -0.03, 0.025, -0.015, 0.02, -0.01, 0.02 and -0.025. Firstly, a single tone varied from 0 to $0.5 F_s$ is tested and the SNDR and SFDR performance are displayed in Fig. 4, which shows an improvement of 30 dB and 45 dB with the proposed calibration method. The wideband (16QAM) signal was also sent to the TI-ADC system as shown in Fig. 5. The SNDR and SFDR are improved from 29.38 dB and 32.63 dB to 61.26 dB and 70.16

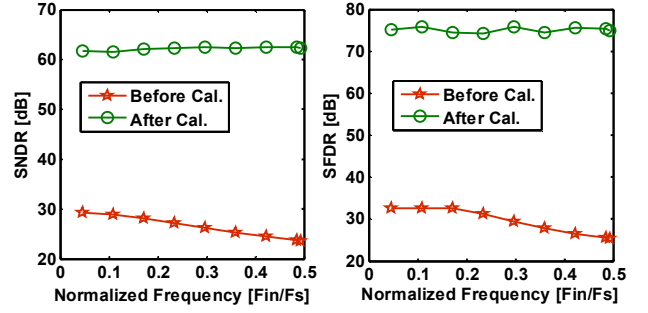


Fig. 4. SNDR (left) and SFDR (right) performance versus the input frequency for a 16-channel TI-ADC.

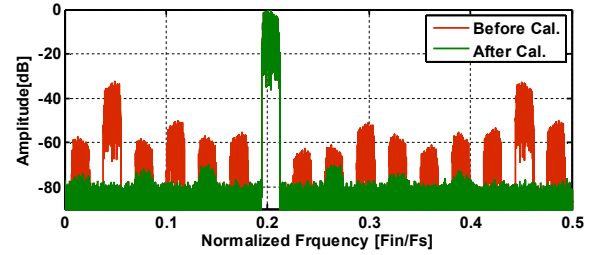


Fig. 5. Spectra of a 16-channel TI-ADC for 16QAM.

dB, which shows an improvement of over 30 dB and 35 dB.

IV. CONCLUSION

This paper has presented a digital blind calibration method to calibrate the gain and time-skew mismatches in any-channel TI-ADCs. Extensive simulations have been presented to verify the effectiveness of the proposed method. Specially, this method could be further modified as an adaptive calibration method based on LMS or RLS algorithm, which will be our future research work.

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